[Samsung PRISM] End Review Report

NEON implementation of mathematical operations for image processing

Multiplication ,Division ,Normalization and Denormalization

Working Team Detail

College Professor(s): Dr.Mallegowda M

Students:

1. Saanvi Nair (1MS21CS110)
2. Sahil Khirwal (1MS21CS111)
3. Vasuman Mishra (1MS21CS133)
4. Manik Animesh (1MS21CS071)

Department: Computer Science and Engineering

Date: January 2025

Work-let Area – AI, ML | NEON implementation of mathematical operations for image processing

SAMSUNG

Work-let duration – 6 months

Problem Statement:

- Implement the following image processing operations done on image buffer
    - Multiplication,
    - Division,
    - Normalization & Denormalization

in NEON & further optimize it in assembly coding

Description:

- Given an image/images of any size, implement multiplication and division of two images in NEON. Also, implement Normalization and Denormalization operations in NEON.
- NDK build set up would be provided which reads the input image using openCV API & save the image to file. This NDK build can be built in windows or linux PC & execution can be done on any armv7 or armv8 samartphone.

References:

1. Normalization Of Image: https://en.wikipedia.org/wiki/Normalization_(image_processing)
2. What is NEON programming - https://developer.arm.com/documentation/102467/0100/Overview
3. NEON documentation – https://developer.arm.com/architectures/instruction-sets/intrinsics/#f:@navigationhierarchiessimdisa=

Mentor:

Akshit Agarwal
Senior Engineer

Rahul Varna
Staff Engineering Manager

Hiren Bharatbhai Gajjar
Architect

Expectations:

- NEON implementation of Multiplication, Division, Normalisation & Denormalization is completed & working for any size of image.
- When image normalisation/denormalization is done, the precision should be maintained as required. Int16 to float16 or float32 to int8.
- Share the execution time of algorithms of NEON code Vs python or c++ code on same smartphone device.
- Optimize the NEON operation using assembly code.
- Image quality assessment
    - OpenCV operations & NEON operations should be similar without any artefacts.
    - Should be done qualitatively & quantitatively (PSNR , SSIM etc)

Training/ Pre-requisites:

- Knowledge of usage of image processing lib openCV on how to read & write image files.

Milestones:

Kick Off < 1st Month >

- Understand the Problem definition & scope of the development.
- Understand the basics of OpenCV APIs to perform all operations – mul, div, norm & denorm.
- Go through the NEON documentation.

Milestone 1 < 2nd Month >

- Implement the operations in python or C++.
- Collect the validation images from smartphone.
- Compare the output of python and C++ with openCV API.
- Start with the implementation of the developed python or c++ code of downscaling into NEON.

Milestone 2 < 4th Month >

- Complete the implementation of atleast one of the algorithms in NEON.
- Generate the outputs of these algorithms on the validation images on python or c++ code & openCV API & NEON.
- Image quality assessment should be done qualitatively (PSNR or SSIM)
- Start the implementation of the assembly code for the above algorithms.

Closure < 6th Month >

- NEON & assembly implementation of all the algorithms are completed.
- A detailed report is prepared with summary of outcome of NEON code with profiling details on the smartphone tested.

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

1. Weighted Threshold Histogram equalisation

Enhances the image contrast by modifying the image histogram by applying weights and thresholds
Weights are calculated based on probability mass function : If the PMF is lower than the threshold, it is set to zero, If it is above the threshold, it is set equal to the threshold.
Otherwise: It is equivalent to (((pmf-low)/(high - low))^r)*high

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

2. BM3D (Block Matching and 3D Filtering)

Two step process applied sequentially. Each step includes 4 steps:
Block matching (Finding nearest similar blocks), Grouping (Grouping the similar blocks based on environment), Collaborative Filtering (Includes 3d transformation using hard thresholding and inverse transformation) and Aggregation (combining the filtered blocks to estimate the denoised image)

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

3. LIME (Low light Image Enhancement)

It is enhancing the image quality using illumination maps. It tells about light condition of the blocks. The illumination map is refined and gamma correction is applied. This is followed by non linear operations for encoding and decoding the luminance (This adjusts the brightness).
The algorithm uses Gaussian kernel affinity to find nearest blocks and smoothen the illumination map. This corrects the under exposure.

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

4. DUAL (Extension of LIME)

LIME algorithm only corrects the under exposure. DUAL algorithm uses the same method to correct the over exposure. It inverts the illumination map of the original image and applies the refinement on the inverse illumination map. After this the illumination map is inverted back to the original form. This corrects the over exposure.
DUAL algorithm corrects the under as well as the over exposure.

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

5. NLM

Non-Local Means is an advanced denoising algorithm that works by averaging similar patches in an image. Unlike local filters, it can preserve fine details while removing noise effectively.

Key Components

1. Patch Comparison
   - Compare neighborhoods around each pixel
   - Weighted average based on patch similarity
   - Gaussian weighting for spatial distance
2. NEON Optimization
   - Vectorized patch comparison (16 pixels at once)
   - Parallel weight computation
   - SIMD-optimized pixel averaging
   - Multi-threaded processing

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

6. Advanced Demosaicing

Demosaicing reconstructs full-color images from Bayer pattern sensor data using advanced interpolation techniques that preserve edges and reduce artifacts.

Key Components

1. Green Channel Interpolation
   - Gradient-based direction selection
   - Edge-aware interpolation
2. Red/Blue Channel Interpolation
   - High-quality linear interpolation
   - Cross-channel correlation
3. NEON Optimizations
   - Vectorized gradient computation
   - Parallel interpolation
   - SIMD-optimized color correction

Work-let Name: NEON implementation of mathematical operations for image processing[Multiplication/Division]

Worklet Details

1. Worklet ID: **24VI14MSRIT**
2. College Name: MS Ramaiah Institute of Technology

Image processing algorithms :

7. Fast Guided Filter

The Fast Guided Filter is an edge-preserving smoothing filter that can process images quickly while preserving edges. It's particularly useful for detail enhancement, HDR compression, and feathering/matting.

Key Components

1. Box Filter Stage
   - Fast mean computation
   - Linear coefficient estimation
2. Guidance Stage
   - Edge-aware filtering
   - Local linear model
3. NEON Optimizations
   - Vectorized box filtering
   - Parallel mean/variance computation
   - SIMD-optimized linear coefficient calculation

Thank you